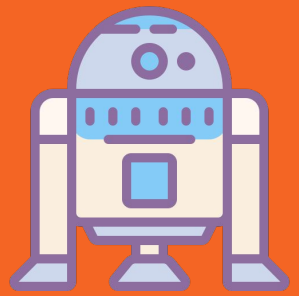

Agentic AI for Healthcare Intake with Classification and Human-in-the-Loop (HITL) Control

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AI in Healthcare High Risk Project

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The Problem: Intake is Critical, But Often Inconsistent

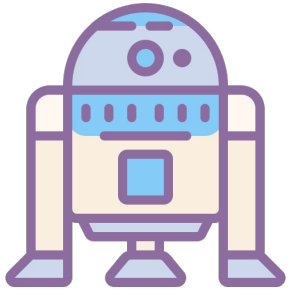
- Patient intake is the critical first step in healthcare but is often variable and error-prone.
- LLMs offer potential for automation, but context window limitations and hallucination pose major challenges
- Current conversational systems risk losing coherence if specialist roles shift mid-dialogue
- Need for flexible systems that combine AI-driven efficiency with human adaptability and oversight

Our Approach: Agentic, Modular, Human-in-the-Loop

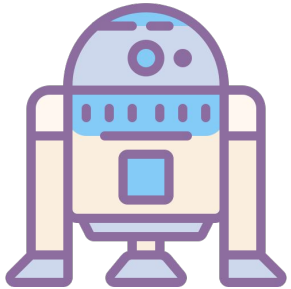
- Built modular AI agents for classification, specialist Q&A, and summarization.
- Assigned specialist roles early via a classifier agent to maintain coherence.
- Embedded human-in-the-loop control at key conversation turns.
- Balanced AI automation with human adaptability for clinical realism

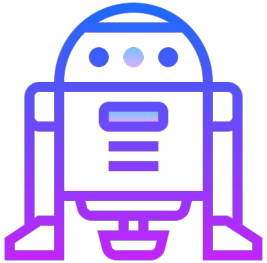
System Design Choice: Classifier Agent

- Early testing showed specialist responses were too generic without knowing the patient's context.
- Added a classifier to predict patient category upfront (e.g., Cardiovascular, Mental Health).
- Classifier output selects targeted specialist prompts, improving relevance and realism.
- Modular structure allows easy expansion (new specialties, new conditions).



Challenges and Trade-Offs





- **Direct Agentic Coding:**
  Manual context chaining gives full control and explainability vs using frameworks like AutoGen.
- **Three Patient Manual Inputs:**
  Early tests showed that 2 turns sometimes missed critical information; 3 inputs create richer case histories.
- **Manual User Inputs vs Synthea Only:**
  Shifted from fully synthetic Synthea data to live manual patient messages to improve realism.
- **Streamlit Implementation:**
  Chose Streamlit for rapid prototyping and testing ease. Lightweight but limited flexibility for complex deployments.

System Design: Step-by-Step Intake Flow

Step 1: User enters initial symptom message.

Step 2: Classifier agent predicts medical category (e.g. Cardiovascular, Mental Health, etc.)

Step 3: Specialist agent assigned.

Step 4: Specialist agent asks follow-up questions

Step 5: User manually enters patient's reply (HITL)

Step 6: Specialist agent asks deeper clarifications

Step 7: User manually enters final patient reply

Step 8: Summarizer agent generates a clinical summary and next steps.

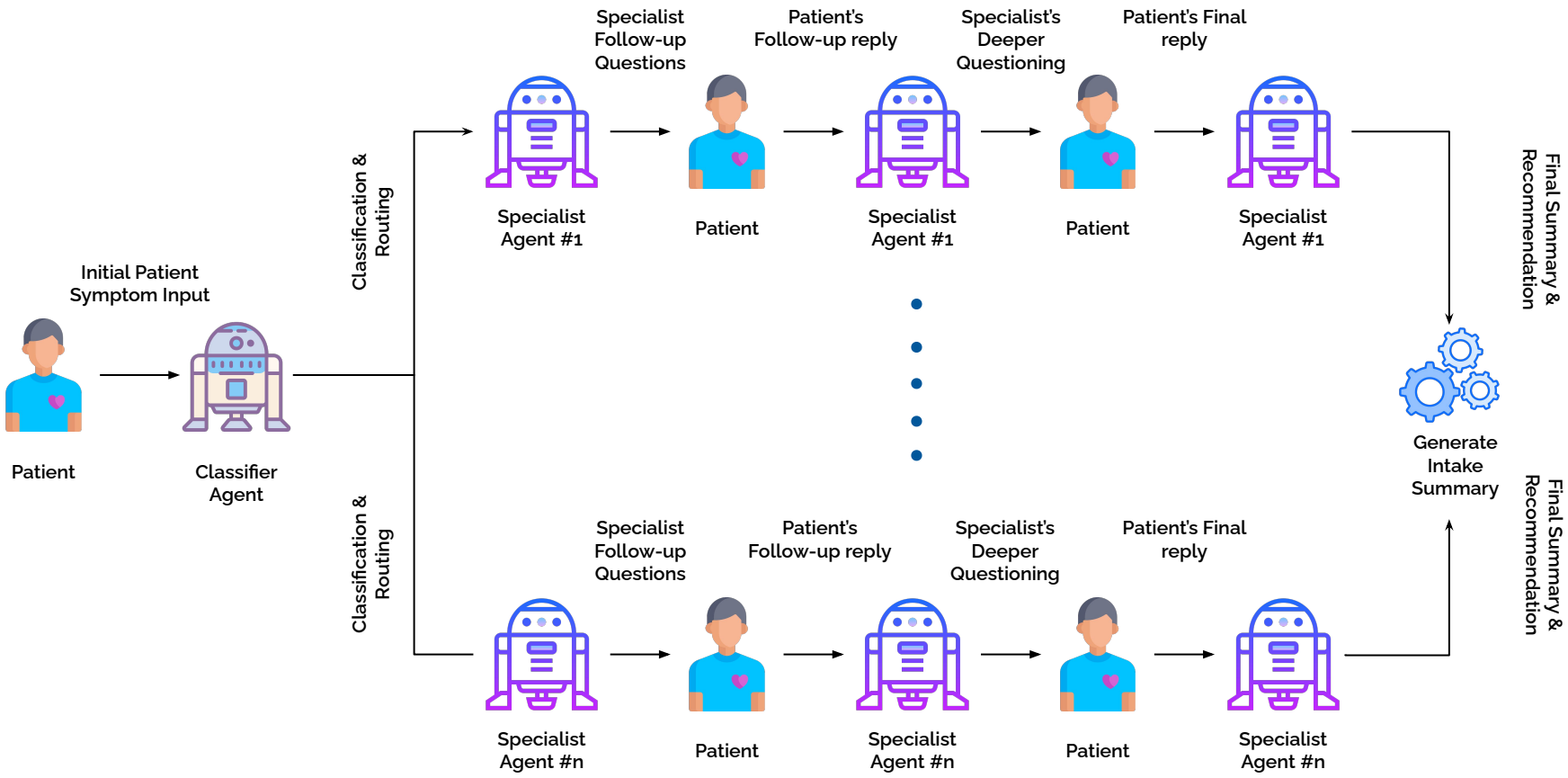


Figure 1. Patient Intake Workflow with a Classifier Agent, Multiple Specialist Agents, and Human-in-the-Loop

Code Walkthrough: Patient Message Classification

- Defines nine intake categories (e.g., Cardiovascular, Mental Health).
- Prompts GPT-4 to classify the patient message into the best-fit category.
- Controls downstream specialist behavior based on classification.
- Uses a low temperature (0.3) to ensure consistent, deterministic outputs.

```
def classify_patient_message(message):
    categories = [
        "Cardiovascular", "Respiratory", "Musculoskeletal", "Mental Health",
        "Endocrine / Metabolic", "Dental / Oral Health",
        "Social / Behavioral", "Medication Review", "Unclear / Needs More Info"
    ]
    prompt = f"""
You are a medical intake assistant. Your job is to classify a patient message
into one of the following categories:

{', '.join(categories)}

Please return **only** the category name and nothing else.

Patient message: "{message}"
"""
    response = client.chat.completions.create(
        model="gpt-4",
        messages=[
            {"role": "system", "content": "You are a medical intake classification
agent."},
            {"role": "user", "content": prompt}
        ],
        temperature=0.3
    )
    return response.choices[0].message.content.strip()
```

Code Walkthrough: Human-in-the-Loop Patient Reply

- After specialist asks first follow-up, user manually enters patient's reply.
- Text input field (st.text_area) captures the human response.
- System records the patient reply and moves to the next specialist question.
- Enables dynamic control and realistic conversation branching.

```
elif st.session_state.step == 2:
  st.markdown(f"** Category:**
`{st.session_state.category}`")
  st.markdown("### Specialist (Turn 2)")
  st.write(st.session_state.turn2)

  patient_followup = st.text_area("Step 2: Enter
Patient's Reply (Turn 3)", key="turn3_input")
  if st.button("Continue to Specialist Follow-Up"):
    if patient_followup.strip():
      st.session_state.turn3 = patient_followup
      st.session_state.turn4 = specialist_followup_2(
        st.session_state.category,
st.session_state.turn1, st.session_state.turn3
      )
      st.session_state.step = 3
      st.rerun()
```

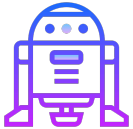
Example: Patient-Specialist Conversation (with Human-in-the-Loop) (Part 1 of 2)

Hello - I am having a headache and a runny nose. I also have some coughing.

I'm sorry to hear you're not feeling well. Could you tell me more about your cough? How long have you been experiencing it and how severe would you say it is? Also, have you noticed if it gets worse in certain environments or situations?

I have had it for about 2 nights now. It has gotten worse actually. I tried taking some over the counter meds like Tylenol, but only works for a few hours.

Can you describe your cough? Is it dry or are you producing any mucus? Also, have you noticed any specific triggers such as certain environments or activities that seem to make your symptoms worse?



My throat hurts when I cough. It is dry I think. I have not noticed any triggers.

Example: Patient-Specialist Conversation (with Human-in-the-Loop) (Part 2 of 2)

You've been experiencing a headache, runny nose, and coughing for the past two nights. Despite taking over-the-counter medications like Tylenol, your symptoms have actually worsened. Your cough is dry and painful, but you haven't noticed any specific triggers that might be causing these symptoms.

Impression: In simple terms, it seems like you might be experiencing a common cold or possibly the flu, as these symptoms are typical of such conditions. However, it's also possible that it could be something else since your symptoms have worsened despite taking over-the-counter medication.

Recommendations:

1. Rest and hydration: Make sure to get plenty of rest and stay hydrated. This can help your body fight off whatever is causing these symptoms.
2. Over-the-counter cough suppressants: Since your cough is dry and painful, you might benefit from an over-the-counter cough suppressant to ease the discomfort.
3. Medical consultation: If your symptoms continue to worsen or persist for more than a week, I recommend you seek medical attention. They may want to run some tests to rule out more serious conditions like a respiratory infection.

Please remember that it's important to listen to your body and take care of yourself. If your symptoms persist, don't hesitate to seek further medical advice. Your health is our primary concern. Please take care.

Key Results: What We Achieved

- Successfully simulated multi-turn patient-specialist conversations across multiple medical categories.
- Embedded human-in-the-loop control at key patient reply stages (Turn 3 and Turn 5).
- System handled both clear and ambiguous patient inputs gracefully (e.g., vague "hi" still led to meaningful clarification).
- Modular agent design allows flexibility to scale or extend (e.g., adding new categories or specialist behaviors).
- Streamlit UI enables real-time testing, prototyping, and user-driven exploration.

Conclusion & Future Work

- Demonstrated a flexible, agentic healthcare intake system combining LLMs and human-in-the-loop interaction.
- Highlighted how modular design enables dynamic, multi-turn conversations across medical categories.
- Future Work:
  - Extend to longer conversations and full patient histories.
  - Add RAG (Retrieval-Augmented Generation) for knowledge-grounded responses.
  - Introduce evaluation metrics (e.g., clinical plausibility, user satisfaction).
  - Explore clinician-facing applications (training tools, triage assistants).